

# SELF-DISTILLED PRONUNCIATION AND ACCENT CONTROL FOR NEURAL TEXT-TO-SPEECH

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## ABSTRACT

Text-to-speech that reads raw text has no lexicon: a rare word is read as guessed. Remedies train a reading-and-accent channel on recorded speech or edit words one at a time from exemplars. We do neither. The frozen backbone reads a sentence containing a *common* word it already says correctly, and its own output then serves as the teacher for the same sentence, with that word replaced by a tagged, accented reading; this training pair is the whole idea. On Sarashina2.2-TTS, screened raters at Fleiss’  $\kappa = 0.85$  hear the prescribed accent on 0.89 of unseen words against 0.57 for kana, which cannot express one; kana wins no pair; naturalness is not measurably hurt. Moved untuned to autoregressive, diffusion, and encoder-decoder backbones, it transfers reading, 0.25 to 0.47 above no edit on 319 words, and on CosyVoice 2 accent on two words in three, but not on Irodori; the paper locates why.

**Index Terms**— speech synthesis, pronunciation control, pitch accent, self-distillation, parameter-efficient fine-tuning

## 1. INTRODUCTION

Conventional cascade text-to-speech (TTS) resolved pronunciation before synthesis: a grapheme-to-phoneme (G2P) front-end consulted a lexicon, and an unknown word was fixed with a lexicon entry. End-to-end TTS trained on raw text dropped that stage, and with it the ability to be *told* how to say a word: a rare kanji, a proper noun, or a domain term is rendered as the model guesses, and the only recourse is to change the model.

For Japanese, the problem is not only which morae are produced but which of them carries the pitch accent. Accent is lexically specified, and getting it wrong is not merely unnatural: the listener searches the lexicon for a word that was never uttered, so the cost falls on recognition.

Input-side control is not new: prosodic symbols in the input sequence were routine in unit-selection and hidden Markov model (HMM) TTS, and Kurihara *et al.* [1] carried them into neural sequence-to-sequence TTS. UtterTune [2] realizes them for a modern LLM-based backbone by wrapping the target word in two special tokens carrying an accented katakana reading. Sarashina2.2-TTS [3] describes the same family of mechanisms (Sec. 3.2.1 of that report). Mixing phoneme or reading input into the training set — CosyVoice 3’s pronunciation inpainting [4], GLM-TTS’s hybrid phoneme-text input [5] — gives control only where the base was trained with it. Training-free editing exists for released models: SonoEdit [6] rewrites one word’s pronunciation in an LLM-based TTS by a closed-form weight edit, and FlowEdit [7] stores a correction as a text-embedding perturbation in a flow-matching TTS; both correct one word at a time from a supplied exemplar (a synthesis from an International Phonetic Alphabet (IPA) transcription, or a

native recording) and address segmental pronunciation, not accent. What is unresolved for a released model is the *cost* of installing the channel afterward and its *reach*. Both published instances of the tag channel pay in recorded speech: 15,097 hand-corrected JSUT/JVS [8, 9] pairs in UtterTune,  $\approx 4,000$  h of supervised fine-tuning in Sarashina2.2-TTS, a capability not included in the released weights.

This paper removes the recording cost entirely. Given a reading spelled out, the model can already say it; what it lacks is a channel that carries the reading and the accent together, and we install that channel by distilling the model against itself. What is new is the training pair, not the adapter or the tag.

**Contributions.** (i) A self-distillation procedure that installs reading and accent control into a frozen TTS backbone using no recorded speech, no human labels, and no corpus — only a text set and a public front-end (§2). (ii) Evidence, from screened raters, that it gives an accent notation a meaning the base never had: on Sarashina2.2-TTS, the tag realizes the prescribed accent on 0.89 of unseen words against 0.57 for plain kana, with kana winning no pair, at no measurable naturalness cost (§4). (iii) The deployment form that makes it usable: register a word once, and every later occurrence is corrected automatically, accent included (§3). (iv) A transfer test: the same recipe moved untuned to three other backbones transfers reading on all three and accent in part on CosyVoice 2 but not on Irodori, and naturalness on none yet; Table 1 lays this out, and the reasons are located (§4.4, §4.6).

**Non-claims.** We claim no novelty for low-rank adaptation or for phoneme-mode tag tokens as such (E2 TTS marks phoneme spans with special tokens [10]), and no language independence; we do not measure sentence-level character error rate (degeneration is measured instead, §4.2), and we do not target homophone disambiguation.

## 2. METHOD

### 2.1. Control signal

The target word is replaced, in the input text, by  $\langle \text{PHON\_START} \rangle$  *accented katakana*  $\langle \text{PHON\_END} \rangle$ . An apostrophe marks the mora after which the pitch falls (the accent nucleus); a slash marks an accent-phrase boundary; an unmarked reading is flat (*heiban*). The notation follows JEITA IT-4006. Across backbones, only the tag spelling changes.

### 2.2. Self-distilled teachers

Let  $w$  be a *common* word that the base pronounces correctly, and  $c(w)$  be a natural carrier sentence containing  $w$  in ordinary orthography. We synthesize  $c(w)$  with the frozen base and keep *its own*

*output* as the target. The student input  $\tilde{c}(w)$  is  $c(w)$  with  $w$  replaced by its tagged accented reading. Distilling

$$\tilde{c}(w) \mapsto \text{base output for } c(w)$$

teaches the model to honor an explicit reading and accent instruction while changing nothing else, because the teacher and the student describe the same utterance. No recording, annotation, or corpus is involved. *Data-free*, as used below, refers to the adapter’s training data; a public G2P front-end supplies the readings, and a public automatic speech recognizer (ASR) is used as a tool where takes (sampled syntheses) are selected (below). How teachers were taken was not uniform, and this is stated. Sarashina and Irodori use ten carriers a word and keep the first take that passes the guard of §4.2, with no selection (4,830 teachers, none dropped) and no reading check; an audit after the fact (every stored teacher decoded and scored with the ASR of §4) hears the target reading in 0.84 (Sarashina) and 0.89 (Irodori) of them; 10 and 15 words are misread in every take, everyday words in uncommon kanji spellings read character by character, and those teachers stay in the set. CosyVoice 2 and T5Gemma use four carriers a word and keep the lowest-CER take against the front-end reading: of three on CosyVoice 2, rejected above CER 0.15 (50 of 1,892 takes dropped, leaving 1,842 teachers over 472 words); of two on T5Gemma, rejected above 0.35, where the base read more than half the carriers too badly for any take to pass (1,068 of 1,932 dropped, leaving 864 teachers over 427 of the 483 words). Selection acts on training data only. T5Gemma’s adapter is thus trained on what an untuned recipe left it, and its results are that recipe’s floor on that backbone, not its ceiling.

### 2.3. The text set, and why odaka shapes it

The word inventory contains 533 words  $\times$  10 carriers, balanced over ten semantic domains, over mora-initial consonant rows, and over all four accent types. One type dictates the design: for *odaka* words, the fall occurs on the *following* particle, so it is inaudible in isolation. Every *odaka* carrier therefore places a particle immediately after the target, and *odaka*, the hardest type, is deliberately over-sampled (15%); because common *odaka* nouns are scarce, a tenth domain (body and person terms), where they are frequent, was added to supply them. Words colliding with any target of the difficult, katakana, or Joyo evaluation sets, whether by surface or by canonicalized reading, are dropped (50 of 533), so the training and evaluation vocabularies are disjoint by construction. The CosyVoice 2 teacher script also excluded the naturalness set’s targets (10 more words); the other three did not, so the naturalness pairs that edit one of those words are excluded (§4.6).

### 2.4. Training

What installs the channel is the training pair in §2: the tagged student input paired with the base’s own output for the plain carrier. The parameterization of the update is incidental: any adaptation that leaves the synthesis stack frozen would serve, up to full fine-tuning of the language model, and we claim nothing for low-rank adaptation. The instance evaluated here, throughout, is rank-16 LoRA [11] ( $\alpha = 64$ ) on the query, key, value, and output projections plus *only the two embedding rows of the tag tokens*, with every other parameter frozen: under 0.5% of parameters move, the flow/vocoder stack is never touched, and training takes about one hour on a single consumer GPU.

Whether the channel can become voice-specific depends on where the speaker enters relative to the adapter. On Sarashina and

T5Gemma, no speaker enters training: the teacher is the base’s own continuation of the plain carrier, and identity is supplied later by the frozen synthesis stack from a reference the adapter never saw. Irodori can also generate without a speaker condition, but its teachers were cycled over a bank of synthetic references. CosyVoice 2 has no such option: its LLM input concatenates a speaker embedding, the text, and the prompt’s speech tokens, so every teacher carries a voice.

**Tag tokens.** Sarashina ships the two tag tokens; on CosyVoice 2 and Irodori, they are added to the tokenizer as two new embedding rows. T5Gemma-TTS’s stock tokenizer splits the string `<PHON_START>` into six subwords, which the decoder then speaks, so two of Gemma’s reserved single-token `<unusedN>` rows serve as delimiters instead, with no vocabulary resize; the accent apostrophe is already a stock token.

## 3. DEPLOYMENT: REGISTER ONCE, APPLIED AUTOMATICALLY

The deployed system maintains a *registry* that maps a surface form to its accented reading — either `{reading, accent index}` or the accented katakana written directly — and applies it via a router at synthesis time.

The router runs the public front-end over the *whole* sentence, matches each registered surface *at morpheme boundaries*, and then

1. wraps only the matched span with the tags, carrying that entry’s accented reading;
2. **absorbs a following suffix** into the wrapper when the front-end attaches one, so that the tag never abuts raw unconverted orthography;
3. **falls back to the unedited base** when no registered surface aligns with a morpheme boundary, and records that the edit did not fire.

Step 2 matters. In distillation, the tag always wrapped a whole front-end token, inflection included. If it instead closes inside one — the stem tagged, its inflection left outside as raw orthography — the model meets a sequence it never trained on, and in our runs it looped on repeated morae (§4.2).

**A word is registered once, with its accent, and every later occurrence in any sentence is corrected automatically**, with no per-sentence markup, no retraining and, unlike per-word editors [6, 7], no exemplar.

## 4. EXPERIMENTS

### 4.1. Setup

We evaluate four backbones spanning generation paradigm, codec, and tokenizer: the autoregressive LLM-based Sarashina2.2-TTS [3] and CosyVoice 2 [12], the diffusion model Irodori-TTS-500M-v3 [13], and the encoder-decoder T5Gemma-TTS [14]. The recipe — text set, LoRA configuration and training procedure — was developed on Sarashina2.2-TTS and applied to the other three without per-backbone tuning; teacher prompting and teacher-take selection differ by backbone as §2 states, and on CosyVoice 2 the number of teacher voices was chosen by a rule fixed in advance (§4.4). Evaluation words are disjoint from training words by construction.

Reading is scored as canonicalized target-word hit under one pass of a kana-constrained ASR (kana-whisper, released with [3]);

**Table 1.** What was tested where, and the outcome. Reading: ASR target-reading accuracy, 319 unseen words, 42 voices (§4.3). Accent: share of words on which screened raters hear the prescribed accent, 50 words, with the comparator judged in the same cell, one registered question per backbone: value over the phoneme-level input (kana), cost against the recorded-data adapter (rec.), transfer over no edit (none) (§4.5). Naturalness NI: non-inferiority at margin 0.15 on the unedited-minus-edited preference (§4.6). “–”: not judged; §4.5 says how rater time was allotted. Sarashina is the backbone on which the recipe was developed; the other three received it untuned.

|                 | Sarashina | CosyVoice 2 | Irodori  | T5Gemma |
|-----------------|-----------|-------------|----------|---------|
| Reading – none  | +.31      | +.47        | +.25     | +.35    |
| Reading – kana  | –.02 n.s. | –.09        | –.07     | –.07    |
| Accent realized | .89       | .66         | .78      | –       |
| comparator      | kana .57  | rec. .96    | none .78 | –       |
| Naturalness NI  | yes –.08  | no +.33     | no +.43  | no +.30 |

the Kana column of Table 2 is the check that the ASR reads each backbone’s output ( $\geq 0.82$  on all four).

**Item sets.** Three sets are used. The 50-word *difficult* set is the listening set. Its ASR reading scores were observed during development, so no reading score is reported; its accent results could not have been tuned to, because no accent judgment existed before the listening study. A 50-word *easy* set per backbone, words whose reading the base already produces, serves the naturalness study alone (§4.6). The second set is drawn from the Joyo Kanji Yomi Benchmark [15], 13,095 native-verified sentences in which one kanji’s reading is marked: we keep every sentence on which a public dictionary G2P (pyopenjtalk [16]) reads the marked span *wrongly*, drop any whose target word or reading occurs in the training inventory, and verify each remaining substitution by reading the substituted sentence back through the G2P against the benchmark’s own pronunciation. Nothing about our models enters the selection, and no sampling is involved: the set is every item that passes, 424 in all. One of four CosyVoice 2 adapters must be chosen (§4.4), and choosing it on the reported items would be selection on the test set, so the 424 items are split *before any adapter is measured* into a 105-item selection half and a 319-item test half by connected component of the (word, reading) graph, so that no word or reading crosses.

**Reference voices.** Zero-shot backbones need a reference prompt, and each ships a different default; measuring each under its own prompt makes the columns incomparable. Every result on the test half, and every listening and naturalness stimulus, is therefore synthesized under one of 42 held-out voices per item, assigned by hash before any audio existed: 42 ReazonSpeech [17] speakers chosen as cluster centroids of a speaker-embedding space, none of which appears in any training or teacher-generation stage. Two ITA-corpus speakers were used only to select among the CosyVoice 2 adapters (§4.4).

#### 4.2. Rejecting degenerate output, identically in every arm

Every backbone occasionally emits non-speech (silence, a repeated mora, a run to the token limit), which a reading metric would score as misses. One guard is applied identically to every arm: a take is rejected if it is under 0.5 s, reaches the generator’s length cap, is silent, is too quiet to hear, or transcribes as fewer than two morae or as one mora repeated six times or more; the seed is bumped, and the *first* valid take is kept. Never the best of several — that would be a

**Table 2.** Target-reading accuracy on 319 unseen rare words (Joyo benchmark items a public dictionary G2P misreads; §4), each under one of 42 held-out voices fixed by hash. Intervals are 95% bootstrap, clustered by word (210 word-reading components), with exact McNemar  $p$ -values. “None” is the unedited input; “Kana” writes the reading in plain kana: the phoneme-level input baseline on the same backbone, which cannot express an accent. UT, the released Utter-Tune adapter, exists only for CosyVoice 2 (15,097 recorded pairs); every “Ours” used no recorded speech; all arms pass the same guard (§4.2).

| Backbone    | None | Kana | UT   | Ours | Ours–Kana, $p$              |
|-------------|------|------|------|------|-----------------------------|
| Sarashina   | .511 | .843 | —    | .821 | –.022 [–.074, +.033], .46   |
| CosyVoice 2 | .279 | .840 | .856 | .749 | –.091 [–.137, –.046], .0003 |
| Irodori     | .586 | .909 | —    | .837 | –.072 [–.110, –.036], .0001 |
| T5Gemma     | .398 | .821 | —    | .752 | –.069 [–.113, –.027], .003  |

**Table 3.** Four CosyVoice 2 adapters differing *only* in the prompts their teachers were generated under, scored on the 105-item selection half under the two selection references A / B (§4); the test half and the 42 voices were not used. Teacher speakers are drawn from one Gaussian; no recorded speech.

| Teacher voices | 1         | 5                | 20        | 20, 4 lengths |
|----------------|-----------|------------------|-----------|---------------|
| Accuracy A / B | .54 / .69 | <b>.68 / .75</b> | .62 / .74 | .63 / .71     |

take selection; the retry count is recorded per item.

Sarashina degenerates on the first take to 0.6% of tagged items, compared with 0.9% in plain kana and 0.9% unedited, and guarding moves its tagged reading accuracy by less than one point (0.828 to 0.821).

Output too quiet to hear (peak below 0.05) is the one failure that redrawing does not cure: it persists in 4.1% of tagged and 2.8% of unedited T5Gemma items, and under 1% on CosyVoice 2, never on Sarashina or Irodori.

#### 4.3. Does control transfer to unseen words?

Table 2 reports reading accuracy on the 319-item test half, words that appear in no training carrier, under one synthesis recipe and one ASR pass for every cell.

Against no edit, the tag gains +0.310, +0.470, +0.251, and +0.354 on the four backbones, with every interval well clear of zero ( $p < 10^{-12}$  throughout): the channel reaches words it never saw tagged in every paradigm. Against plain kana, it reads 0.069–0.091 lower on three of the four ( $p \leq 0.003$ ), one word in eleven to fourteen, which §5 names as the open problem, and is indistinguishable on Sarashina (–0.022, [–0.074, +0.033]).

Voice matters: over the 150 pairs of test items sharing a word and reading under different voices, renditions disagree more often under the tag than with no edit on all four backbones (Sarashina 26.0% vs. 16.0%, CosyVoice 2 28.7% vs. 14.0%, Irodori 18.0% vs. 13.3%, T5Gemma 29.3% vs. 22.0%).

SpeechSplice [18], concurrent work, edits pronunciation on T5Gemma-TTS with no training by replacing encoder hidden states; the present method needs no such intervention point.

#### 4.4. What does zero recorded data cost, and when?

CosyVoice 2 is the one backbone with a released UtterTune adapter, trained on 15,097 hand-corrected recorded pairs. On the 319-item test half under 42 voices the data-free adapter trails it by 0.107 ( $[-0.160, -0.056]$ ,  $p = 8 \times 10^{-5}$ ; Table 2), and on accent by 0.30 (§4.5).

One cause is visible in how a teacher is produced. Sarashina emits semantic tokens from text alone, so its teachers carry no speaker. CosyVoice 2 cannot emit audio without a reference clip, so every teacher is spoken in some voice, and the prompt speaker is baked into the target’s pronunciation. Table 3 varies only the number of synthetic teacher voices, on the selection half under two references used for this choice alone. The choice is a worst case over references, so it uses two clean ones rather than many: ITA-corpus [19] studio recordings, one female (A = amitaro) and one male (B = kamijou), held out from every training and teacher stage and from JSUT/JVS, so a drop under either is the adapter’s, not the clip’s.

Diversity is necessary but saturates: one voice to five lifts reference A from 0.54 to 0.68 ( $p = 0.020$ ), while twenty buy nothing over five ( $p = 0.33, 1.00$ ). The five-voice adapter was chosen by a rule fixed before the numbers existed, the higher of the two references as the minimum, and is used throughout.

Reading accuracy is not where this method wins (Table 2, §4.4); what none of it settles is accent, because kana cannot express one, and writing the accent into the kana does not help: to the frozen base, an apostrophe is punctuation, not a nucleus. Giving that notation a meaning is what the distillation does, and §4.5 measures it.

#### 4.5. Listening evaluation

Accent is judged by ear. Rater time goes to the two backbones where the recipe ran as designed and to the one with a recorded-data comparator; T5Gemma, whose channel was trained on the half of its teacher set that survived (§2), is scored objectively only. Screened raters judge 473 complete sentences, blind and in randomized order, each under one of 42 held-out voices, fixed by hash, so each cell spans 27–31 voices.

Accent realization is a correctness judgment, so ability is gated by a known-answer screening test rather than by credentials. Admission requires 18 of 19 known-answer items, a threshold fixed before recruitment; the first attempt is binding. Candidates work with Japanese pitch accent, either professionally or in training; every recruitment route takes the same test, every candidate is paid according to their score, and no rater is told which rendition comes from which system.

Each admitted rater judges all 473 stimuli plus eight known-answer catch trials. Fleiss’  $\kappa$  is reported *before* any system comparison; the bootstrap resamples words and raters; duplicate items give within-rater consistency, and a rater below 6/8 on the catch trials is excluded and reported. Three contrasts are confirmatory (McNemar exact, Holm-corrected): the edit against no edit on Sarashina2.2-TTS and on Irodori, and on Sarashina, the edit against *kana input*, the contrast on which the method’s value rests (§4.4). The design and analysis were fixed before any audio was heard. The stimulus and screening definitions, the evaluation scripts, the text set, the teacher sets, the adapter weights (under each backbone’s license), and the training code are released.

**Results.** Thirty-two candidates were screened (16 from an open call, 15 by individual contract, 1 by direct approach) and five passed (2, 2, 1); one could not be reached, so four raters were admitted, all sessions complete. Fleiss’  $\kappa$  on the binary outcome (prescribed

accent realized or not), over the 448 stimuli left after excluding duplicates and “cannot judge” answers, is 0.85, pairwise Cohen’s  $\kappa$  0.83–0.87; catch-trial scores are 6–8 of 8 and duplicate consistency at least 22/23, so no rater is excluded; 4 tied items are excluded. On Sarashina, the tag realizes the prescribed accent on 0.89 of words against 0.57 for kana input (risk difference +0.32 [+0.19, +0.45], Holm  $p = 2 \times 10^{-4}$ ), and kana wins no pair (discordant 0 vs. 15); against no edit it is 0.88 vs. 0.66 (+0.22 [+0.06, +0.38], Holm  $p = 0.038$ ). Over no edit, the gain is reading: no edit draws 61 reading and 9 accent failure judgments, the tag 1 and 26, and with reading failures removed, the two are level (0.89 vs. 0.94,  $p = 0.69$ ). Over kana, the gain is accent alone. On Irodori, the tag and no edit tie at 0.78 (+0.00 [−0.16, +0.16]). Editing a *different* word in the sentence moves the judged word’s accent from 1.00 to 0.94 ([−0.14, 0.00]). On CosyVoice 2, the data-free adapter realizes the accent on 0.66 of words against 0.96 for the adapter trained on 15,097 recorded pairs (−0.30 [−0.44, −0.16]).

#### 4.6. Naturalness

Following the critique of absolute rating scales [20, 21], listeners choose the more natural of the *same sentence* with and without the edit, or no difference, on the easy set (§4; 197 pairs, one held-out voice each), where the edit does not change the word heard.

**Results** (86 sittings, 76 complete, 3,228 judgments on 168 pairs, 15–22 per pair). Excluded are one account’s second sitting, three sittings answered faster than the clips could be played (median interval under 4.7 s), and the 29 pairs whose edited word was in that backbone’s training inventory (10, 10 and 9 on Sarashina, Irodori and T5Gemma; §2). With ties split evenly, the share preferring the unedited rendition minus the share preferring the edited one is −0.08 [−0.24, +0.08] on Sarashina and +0.43 [+0.31, +0.56] on Irodori, +0.33 [+0.19, +0.46] on CosyVoice 2 and +0.30 [+0.11, +0.48] on T5Gemma. The registered non-inferiority margin of 0.15 is met on Sarashina, the backbone on which the recipe was developed, and on none of the three was it moved to untuned. The 127 pairs in which both renditions yield the same reading give the same picture (−0.07, +0.41, +0.36, +0.34): the cost lies in how the word is said, not in which word.

## 5. CONCLUSION

A pronunciation and accent control channel can be installed into a frozen TTS backbone for the price of a word list and one GPU-hour, with no recorded speech: distilling the model against itself on words it already says correctly is enough for the control to reach words it never saw, and a registry applies it wherever a registered word appears.

Table 1 says where the channel is worth having. Reading transfers on all four, within 0.02–0.09 of plain kana. Accent was judged on three: on Sarashina2.2-TTS, the tag delivers it where kana cannot (0.89 vs. 0.57); on CosyVoice 2, it reaches two words in three, compared with the recorded-data adapter’s 0.96; on Irodori, it exceeds no edit. Naturalness is non-inferior on the backbone the recipe was developed on, and on none of the three was it moved to an untuned state. The mechanism is the point: a model that already produces the behavior can be taught to request it from its own output alone. Where the recipe was developed, the channel carries reading and accent at no cost; moved untuned, reading everywhere and part of the accent; the rest is next. **Limitations.** One language; four raters; one recipe moved untuned; uneven teacher takes.

## Funding and conflicts of interest

This work received no external or institutional funding. Compute was performed on the author’s own hardware, and the listening studies were paid for by the author personally. The author declares that the method evaluated here is their own, which is why every reported accent judgment is made by raters admitted through a known-answer screening test, none of whom is told which rendition comes from which system, under a design fixed before any audio was heard. One panel rater was approached directly by the author; screening outcomes are reported by route (§4.5).

## Compliance with ethical standards

The listening studies involve non-invasive listening judgments on synthesized speech. No audio, biometric, or personally identifiable data is collected from participants; only their judgments are recorded. All participants receive a written description of the task, the use of the data, and the compensation, and give informed consent before taking part. Participation is voluntary and may be discontinued at any time without penalty; a discontinued session is still paid. Raters are recruited through a commercial crowdsourcing platform and, in one case, through a direct approach; every candidate takes the same qualification screening, and those who do not pass are compensated for it regardless. Raters are asked to pause every 25 minutes and may stop and resume at any time.

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